Motion Center Initial Medical History Report
Shriners Hospitals for Children - Northern Cal

Date: Pre-Op
Age: 10y 1m

DIAGNOSIS

Primary: Cerebral Palsy
Secondary: Spastic
Pattern: Right Hemiplegia Side Affected: Right
Etiology: Evaluation with magnetic resonance brain imaging identified abnormalities in the left cerebral hemisphere as manifested by periventricular subcortical heterotopic gray matter and polymicrogyria, most pronounced in the left frontal lobe perirolandic region.

FAMILY INFO

Primary Care Provider: Father; Mother Siblings: 2

BIRTH HISTORY & DEMOGRAPHICS

Gender: Male Delivery: Vaginal Time in Hospital: WNL
Gestational Age: Full Term Birth Weight: 7lb 2oz
NICU: No Mechanical Vent: No Seizures at Birth: No
Pregnancy Problems: None

DEVELOPMENTAL MILESTONES

Motor: Walk: 12mos. Cognitive Status: Normal
School: Level: 5 How: Mainstream w/ Resource
Activities: Computer: Yes Sports: Basketball

AMBULATORY STATUS

GMFCS: 1 Last 6-12 Months: Worse
Orthotics: Left: None
Right: None
Trips: 1 per day Falls: 0 per
Wheelchair Ever Used: No
Endurance (FMS): 5m (house): 6 50m (school): 6 500m (mall): 6
Endurance: Unlimited
Occupational Therapy: None
Physical Therapy: None
Other Therapy Notes: Previously had PT 2x per week a year ago
Problems/Complaints: POC reports stiff right ankle, right hip higher than left, flexed right knee, and right heel never touches the ground.

MEDICAL HISTORY

Seizures: No
Behavior/Attn Deficit: No
Comorbidities:
Allergies: Seasonal
Medications: Allergy medication

PAST LOWER EXTREMITY ORTHO/NEURO SURGERY

Motion Center Physical Exam Report – Page 1
Shriners Hospitals for Children - Northern Cal

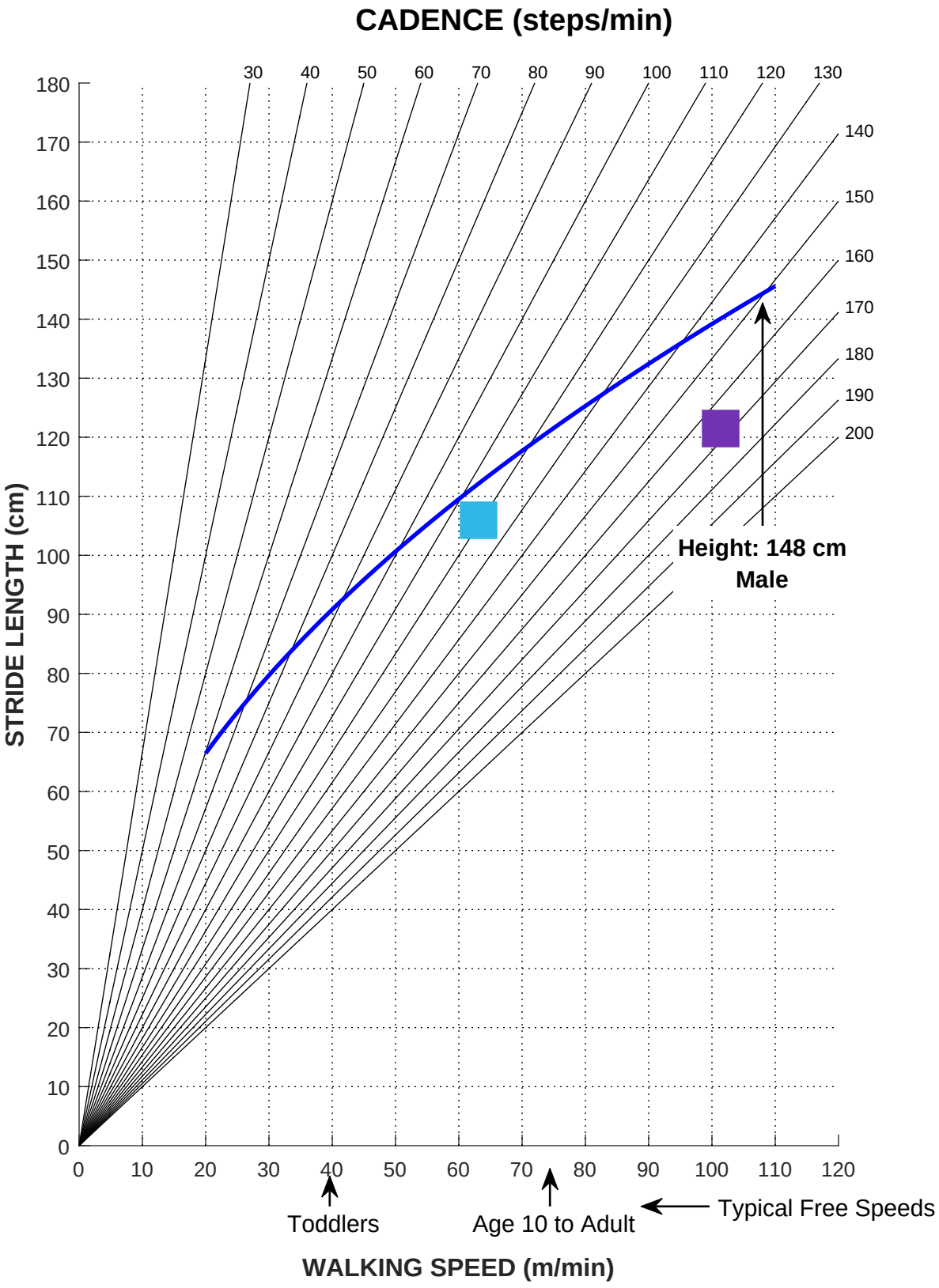
[#] = Typical Development

		Pre-Op	
TRUNK			
	Abdominal Strength	5	
	Spine Straight	Yes	
HIP		Left	Right
ROM	Flexion° [115-125]	125	125
	Extension° (Thomas) [0-15]	0	0
	Abduction w/ Ext° [45]	45	40
	Adduction [30]	25	30
	Internal Rotation° [30-40]	60	75
	External Rotation° [45]	35	10
	Ober Test	NT	NT
Strength	Flexion [5]	5	5
	Extension [5]	5	4
	ABduction [5]	5	5
	ADduction [5]	5	5
KNEE		Left	Right
ROM	Flexion° (supine) [125-135]	160	160
	Flexion° (prone, R1)	NT	NT
	Flexion° (prone, R2) [125-135]	NT	45
	Ely Test	-	+
	Extension° [0]	0	0
	Extensor Lag° [0]	0	10
	SLR° [80-90]	45	40
	Popliteal Angle° [20-0]	40	60
	Patella Alta [no]	NT	NT
Strength	Flexion [5]	5	5
	Extension [5]	5	4
ANKLE/FOOT		Left	Right
ROM	Plantarflexion° [45]	45	60
	Dorsiflex w/ Ext° [10-15]	10	-15
	Dorsiflex w/ Flex° [15-20]	20	-15
Strength	Plantarflex [5]	5	2+
	Dorsiflex [5]	5	5
	Inversion [5]	5	3
	Eversion [5]	5	3
SELECTIVE CONTROL (SCALE)		Left	Right
	Hip [2]	2	2
	Knee [2]	2	2
	Ankle [2]	2	1
	Subtalar [2]	2	1
	Toes [2]	2	1
	SCALE Total [10]	10	7
CONFUSION TESTS		Left	Right
	Unresisted	-	-
	Resisted (classic)	-	+Neutral
	Active Dorsiflexion	Neutral	Everted
CLONUS		Left	Right
	Posterior Tibialis	-	-
	Gastroc-Soleus	-	+Unsustain

Motion Center Physical Exam Report – Page 2
Shriners Hospitals for Children - Northern Cal

		Pre-Op Left	Pre-Op Right
STATIC ALIGNMENT		Left	Right
Standing	Pelvic Obliquity [level]	Down	Up
"	Knee (coronal) [neutral]	Neutral	Neutral
Supine	Leg Length (cm)	78	77
Prone	Femoral Anteversion°	20 AV	30 AV
"	Thigh-Foot Angle°	5 Ext	10 Ext
Supine	Transmall Axis knee ext	20 Ext	15 Ext
Prone	Transmall Axis knee flex	20 Ext	20 Ext
STATIC FOOT (non-wt-bear)		Left	Right
	Hindfoot Class [neutral]	Varus	Varus
	Arch Class. [neutral]	Decreased	Normal
	Forefoot Class. [neutral]	Supinate	Neutral
	Great Toe Class [neutral]	Neutral	Neutral
	Flexibility [flexible]	Flexible	Stiff
STATIC FOOT (weight-bear)		Left	Right
	Hindfoot Class [neutral]	Valgus	Equinus; Neutral
	Arch Class. [neutral]	Absent	Decreased
	Forefoot Class. [neutral]	Pronate	Neutral
	Great Toe Class [neutral]	Neutral	Neutral
CALLOUSES (non-wt-bear)		Left	Right
	Heel	Normal	Absent
	Lateral MTH	Decreased	Decreased
	Medial MTH	Increased	Increased
	Great Toe	Normal	Normal

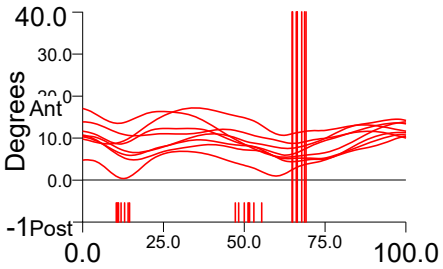
Comments: Right pelvis downward alignment secondary to contracture.



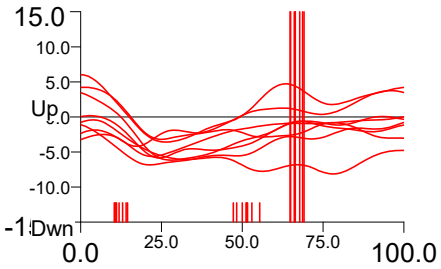
Date/Trial	Velocity (m/min)	Stride Length (cm)	Cadence (steps/min)
PreOp Bare walk	63 (m/min); 89%N	106 (cm); 90%N	119 (steps/min); 99%N
PreOp Fast walk	101 (m/min); 143%N	121 (cm); 103%N	167 (steps/min); 139%N

Right Kinematics Consistency

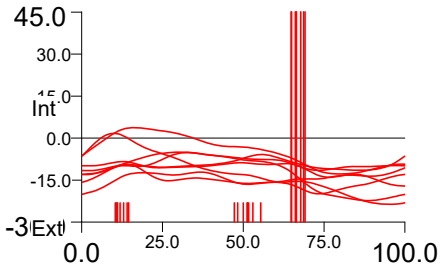
Trunk Tilt



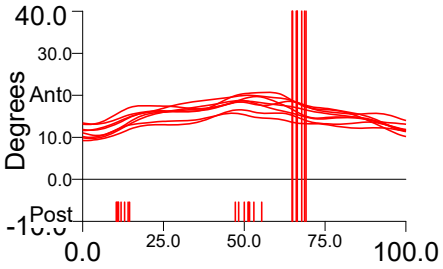
Trunk Lateral Bend



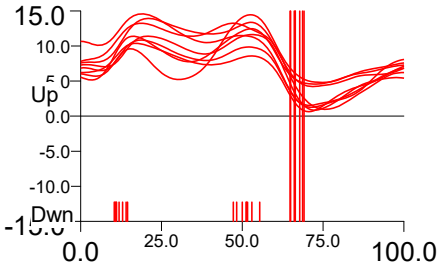
Trunk Rotation



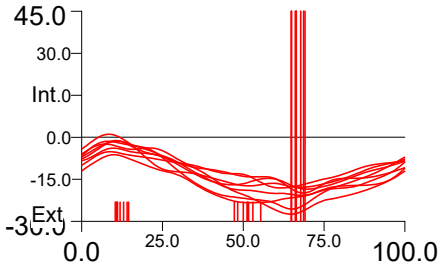
Pelvic Tilt



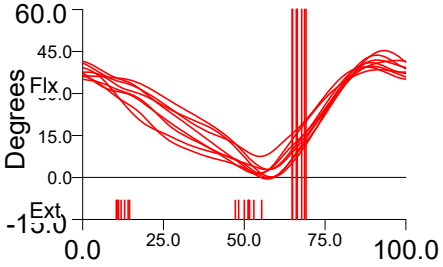
Pelvic Obliquity



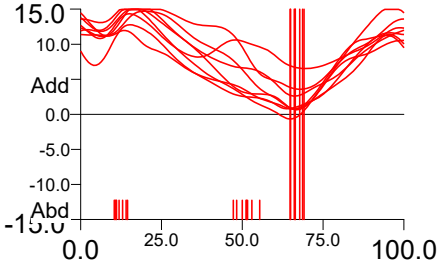
Pelvic Rotation



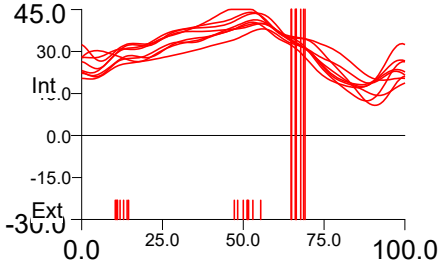
Hip Flexion



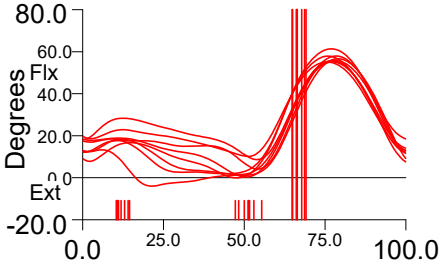
Hip Ab/Adduction



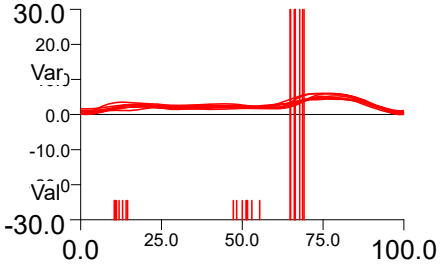
Hip Rotation



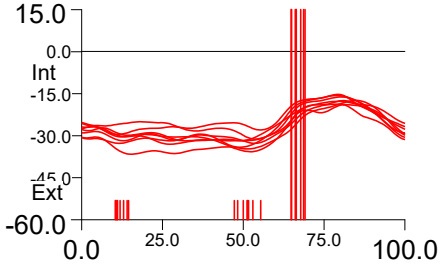
Knee Flexion



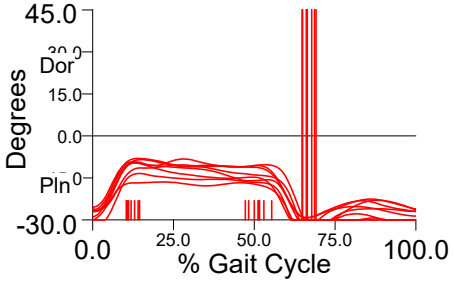
Knee Var/Valgus



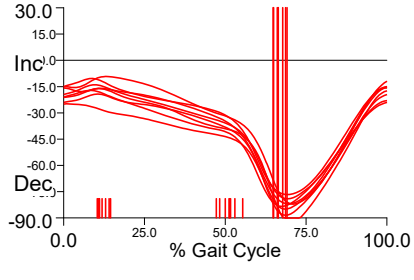
Knee Rotation



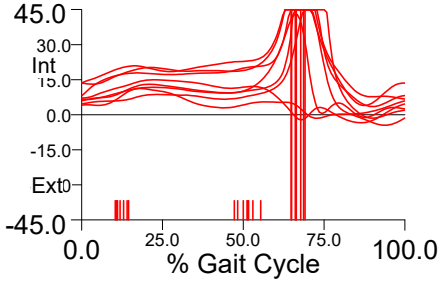
Ankle Dorsiflexion



Foot Inclination



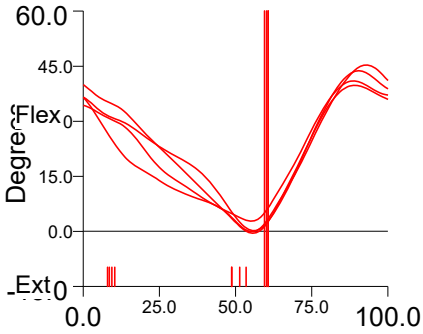
Foot Progression



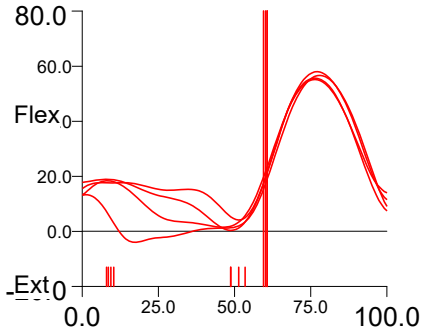
SHCNC - PreOp

Right Sagittal Kinetics Consistency

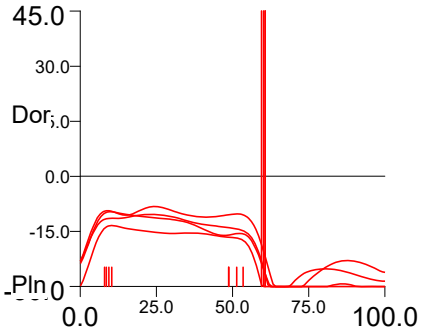
Hip Flexion



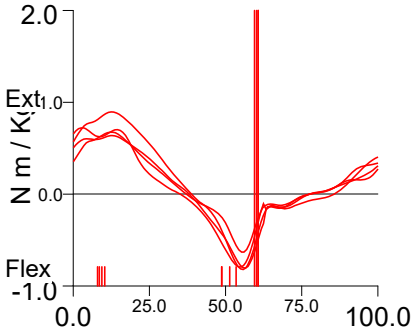
Knee Flexion



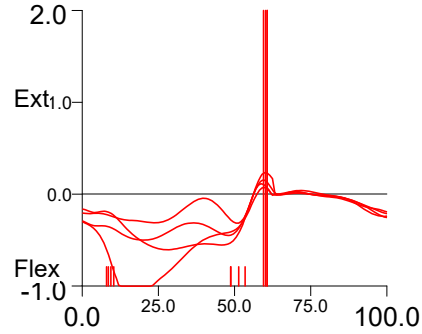
Ankle Dorsiflexion



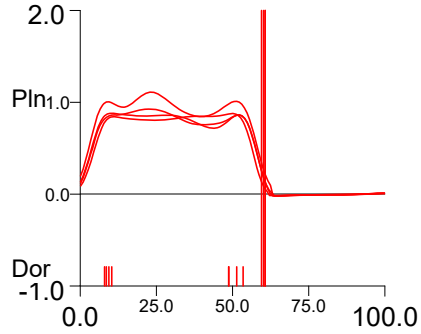
Hip Moment



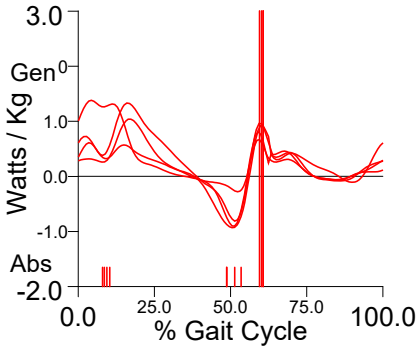
Knee Moment



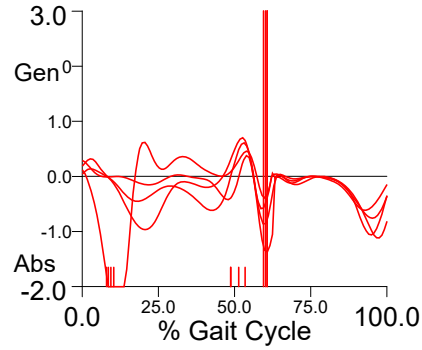
Ankle Moment



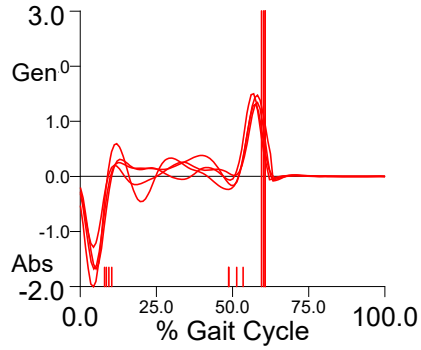
Hip Power



Knee Power



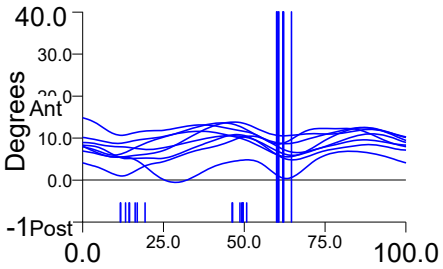
Ankle Power



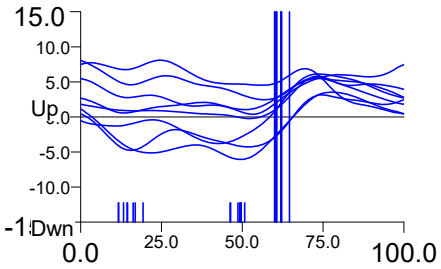
SHCNC - PreOp

Left Kinematics Consistency

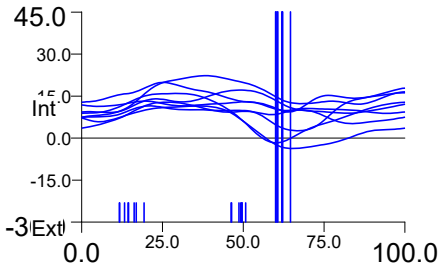
Trunk Tilt



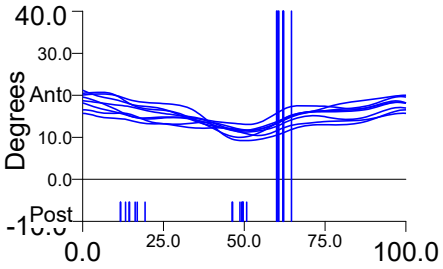
Trunk Lateral Bend



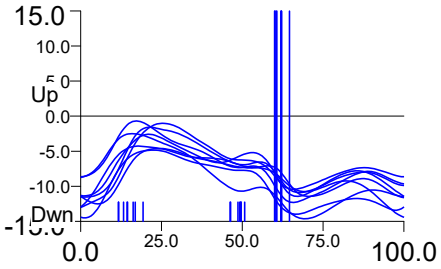
Trunk Rotation



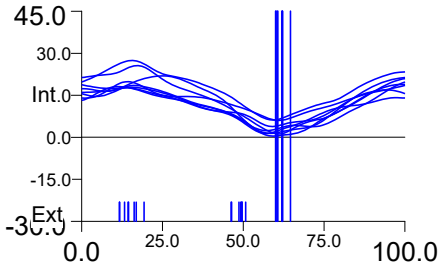
Pelvic Tilt



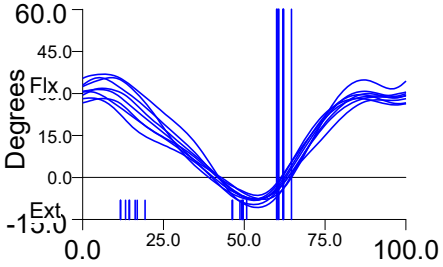
Pelvic Obliquity



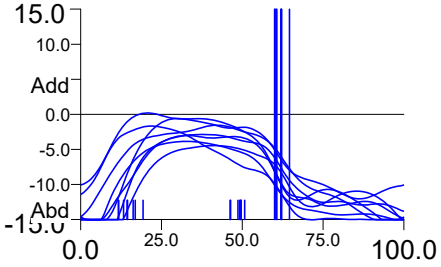
Pelvic Rotation



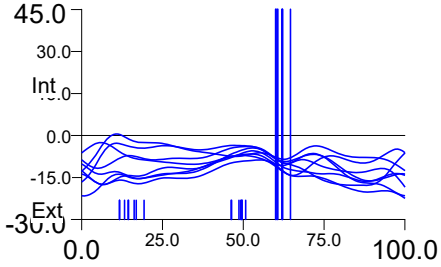
Hip Flexion



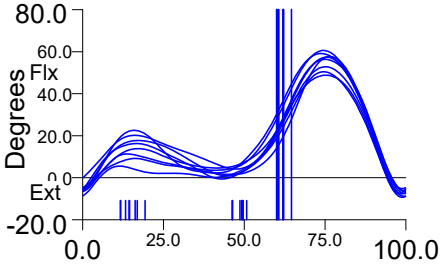
Hip Ab/Adduction



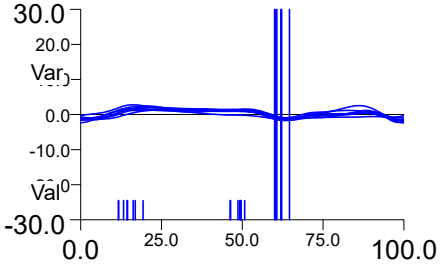
Hip Rotation



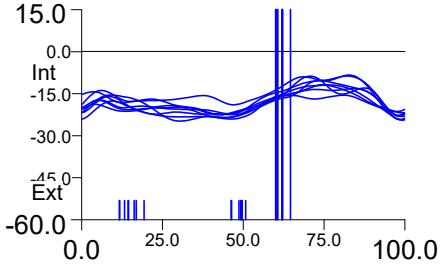
Knee Flexion



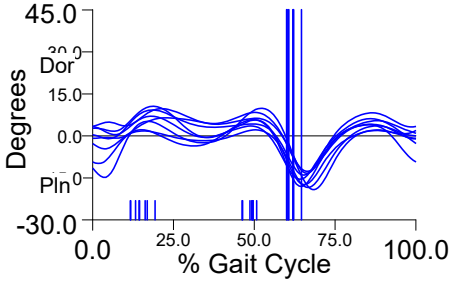
Knee Var/Valgus



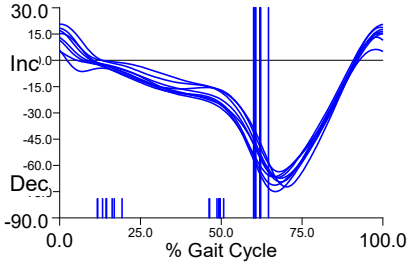
Knee Rotation



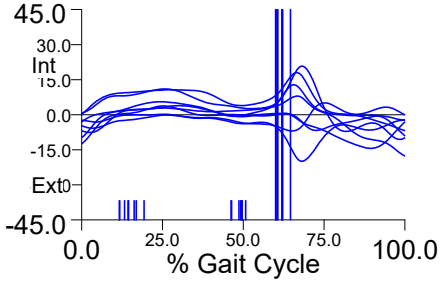
Ankle Dorsiflexion



Foot Inclination

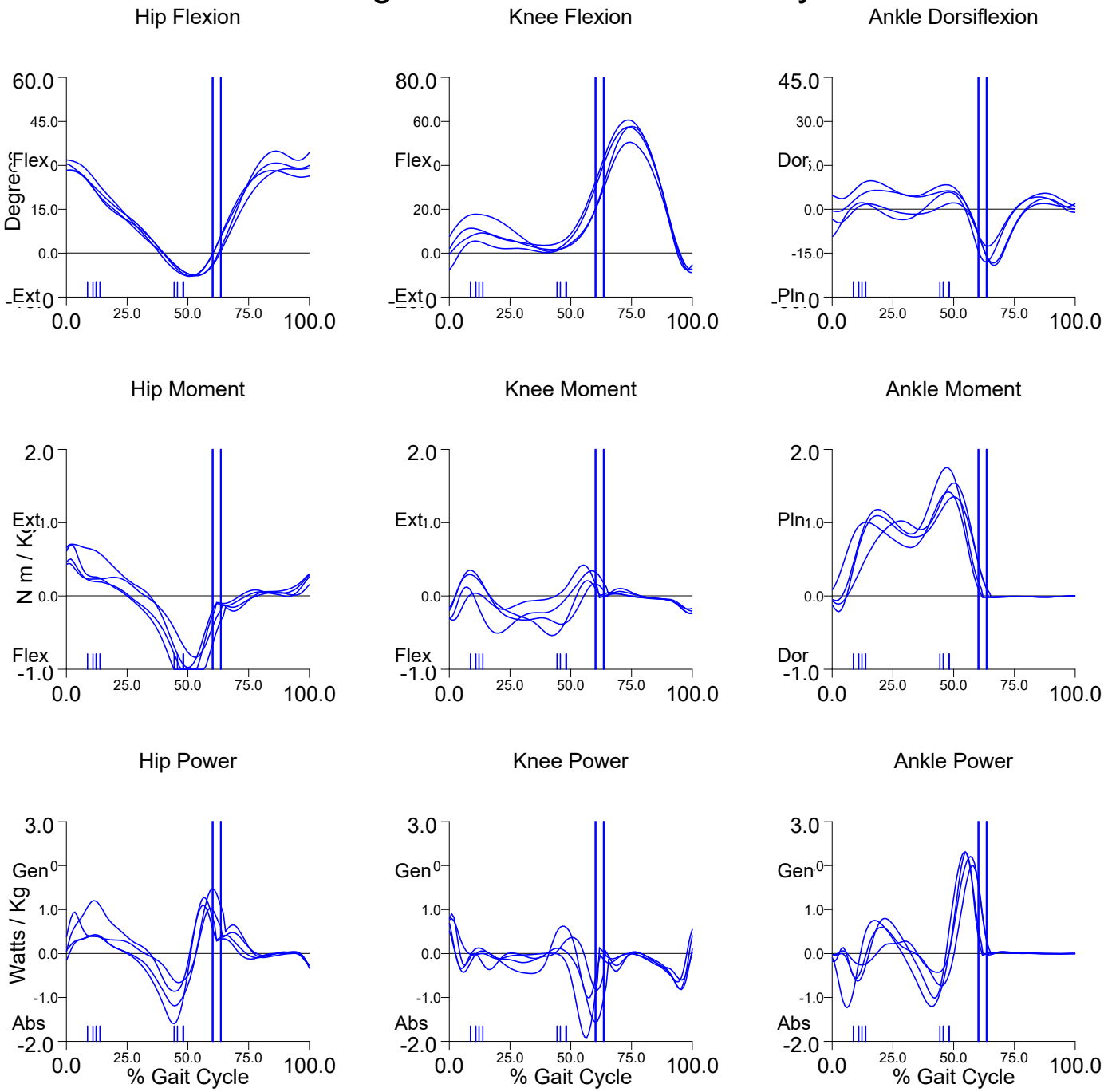


Foot Progression



SHCNC - PreOp

Left Sagittal Kinetics Consistency



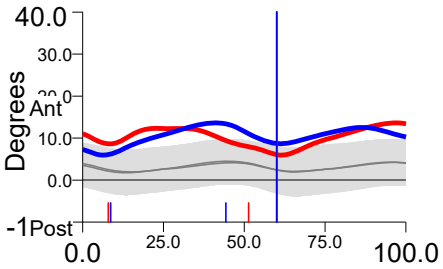
SHCNC - PreOp

RIGHT ---

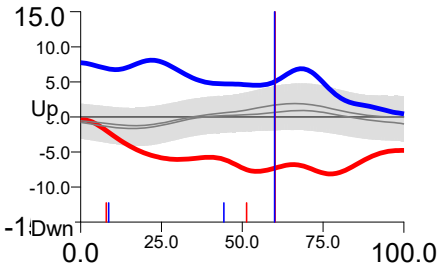
Kinematics Representative

LEFT ---

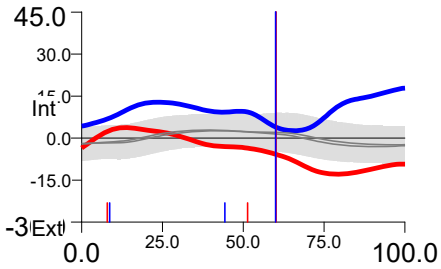
Trunk Tilt



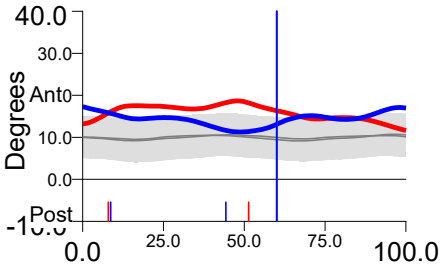
Trunk Lateral Bend



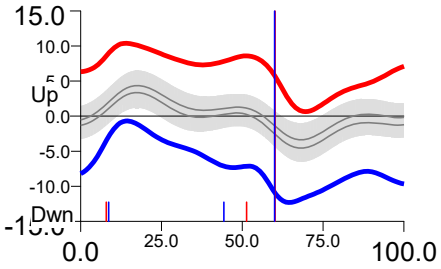
Trunk Rotation



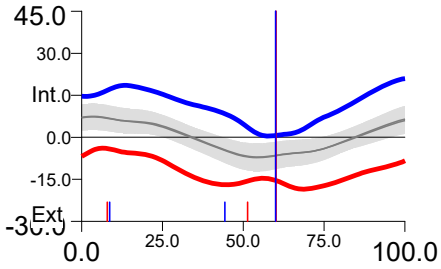
Pelvic Tilt



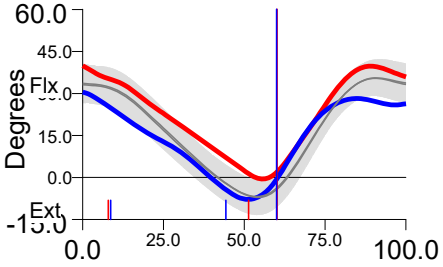
Pelvic Obliquity



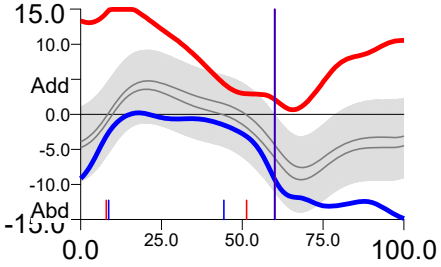
Pelvic Rotation



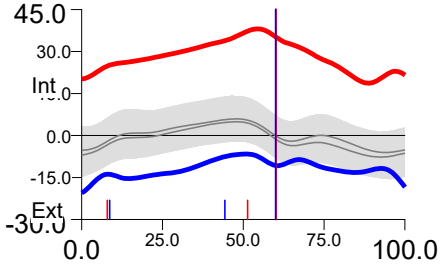
Hip Flexion



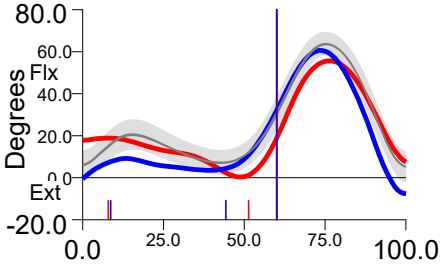
Hip Ab/Adduction



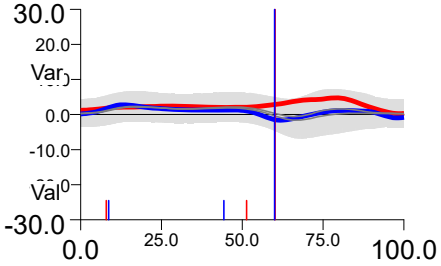
Hip Rotation



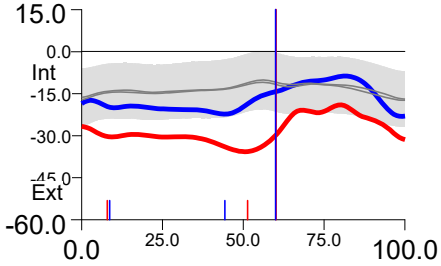
Knee Flexion



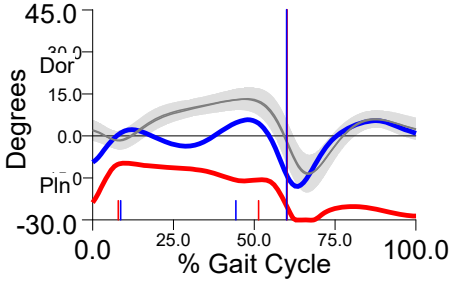
Knee Var/Valgus



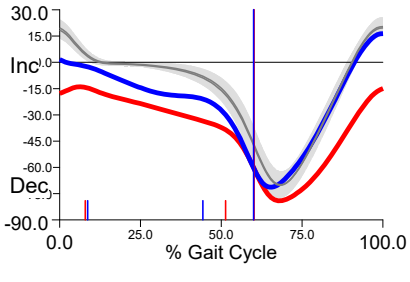
Knee Rotation



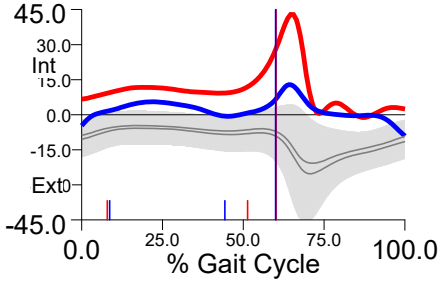
Ankle Dorsiflexion



Foot Inclination



Foot Progression



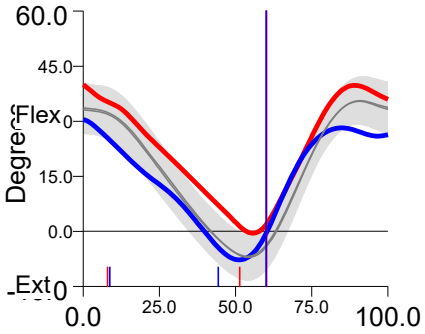
SHCNC - PreOp

Sagittal Kinetics Representative

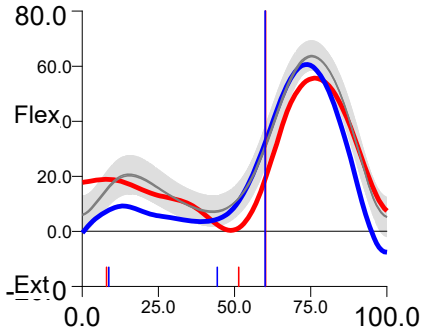
RIGHT ---

LEFT ---

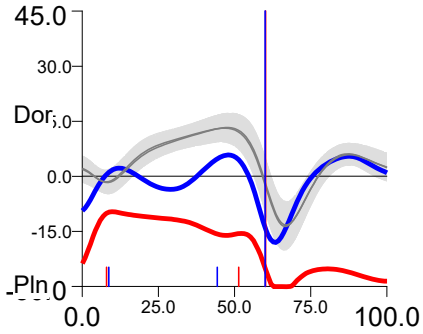
Hip Flexion



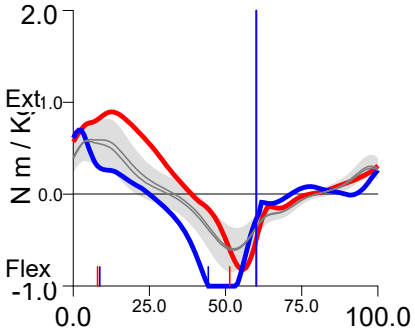
Knee Flexion



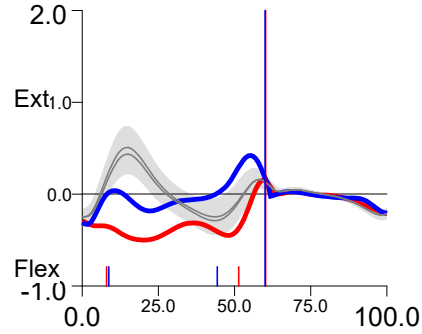
Ankle Dorsiflexion



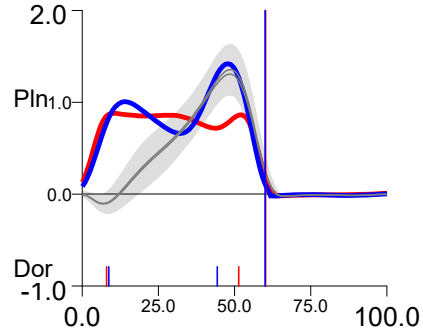
Hip Moment



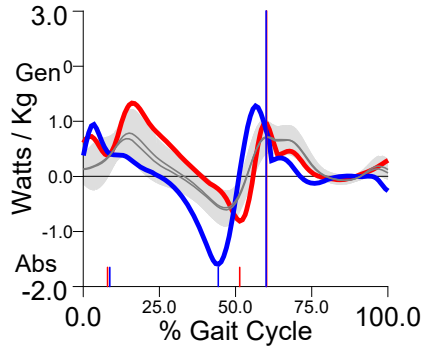
Knee Moment



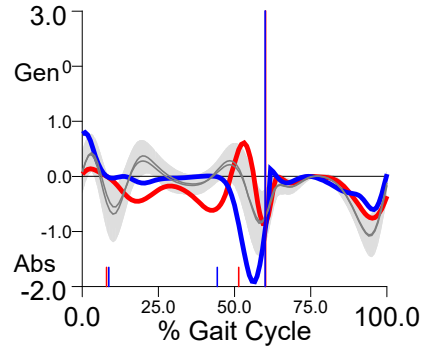
Ankle Moment



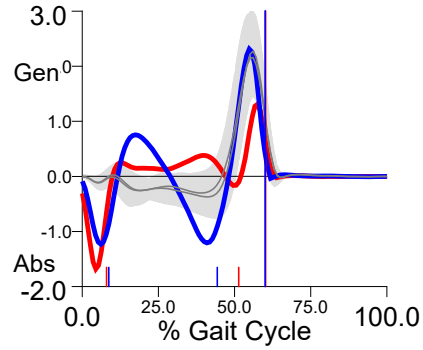
Hip Power



Knee Power



Ankle Power



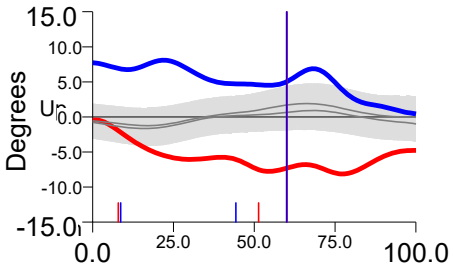
SHCNC - PreOp

Coronal Kinetics Representative

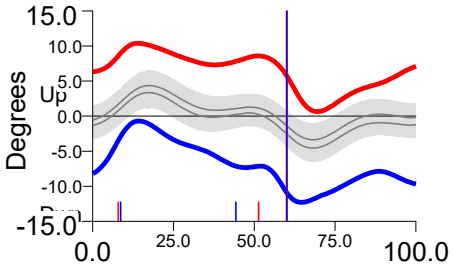
RIGHT ---

LEFT ---

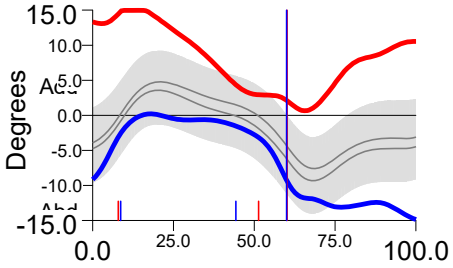
Trunk Lateral Bend



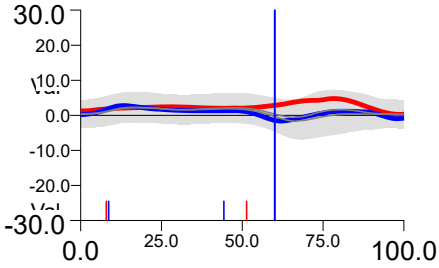
Pelvic Obliquity



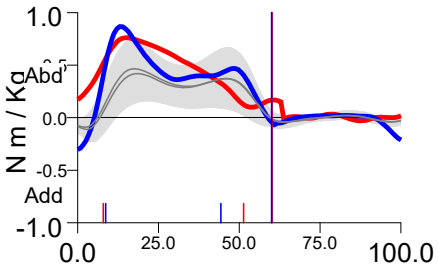
Hip Ab/Adduction



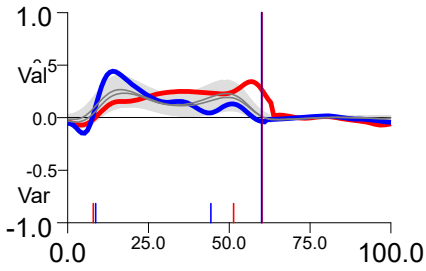
Knee Var/Valgus



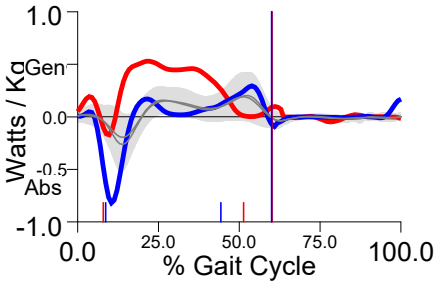
Hip Moment



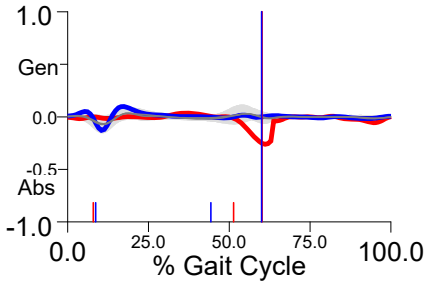
Knee Moment



Hip Power



Knee Power

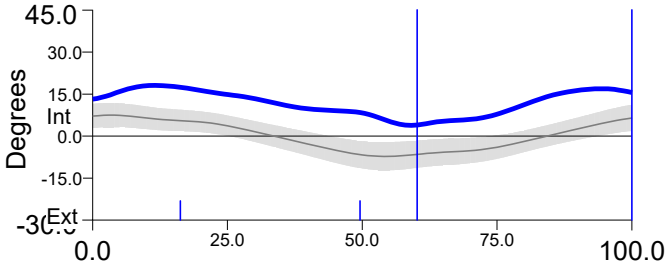


SHCNC - PreOp

Rotational Profile

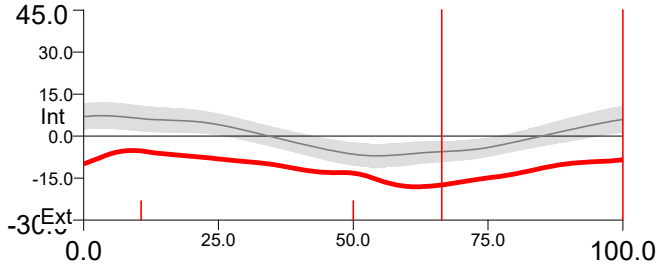
LEFT ---

Pelvic Rotation

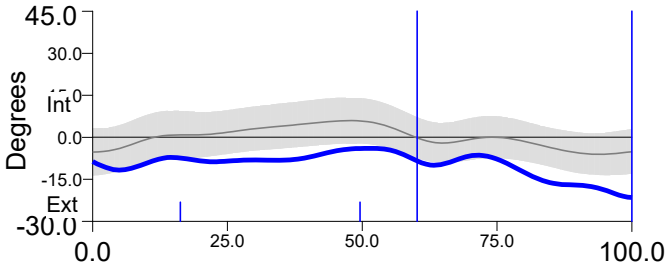


RIGHT ---

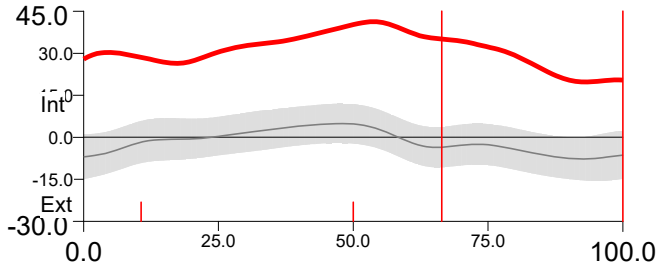
Pelvic Rotation



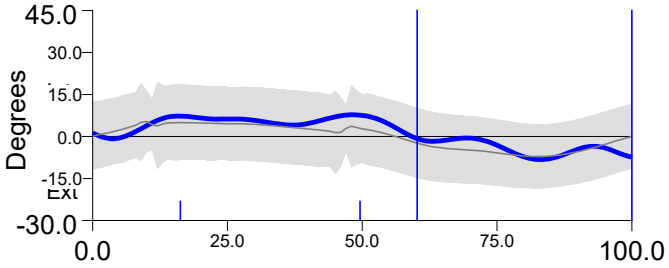
Hip Rotation



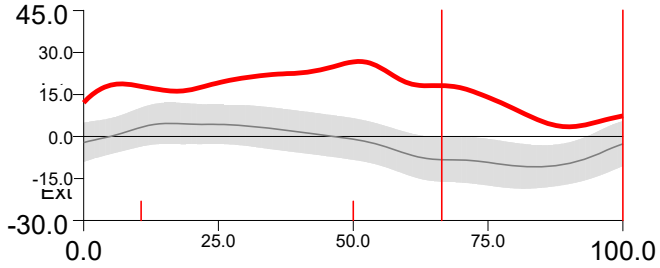
Hip Rotation



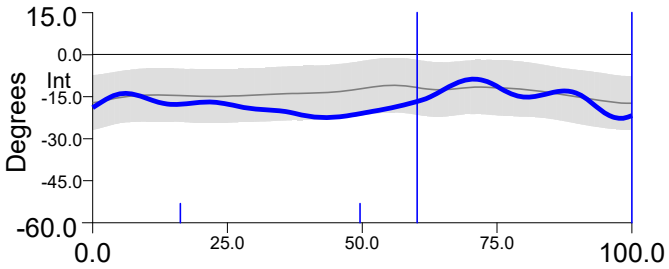
Knee Progression



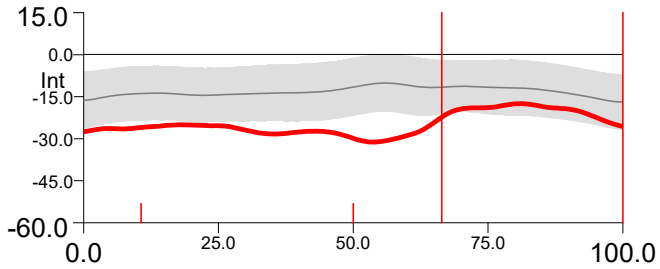
Knee Progression



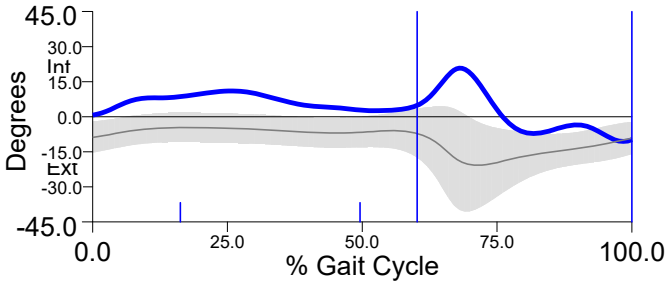
Knee Rotation



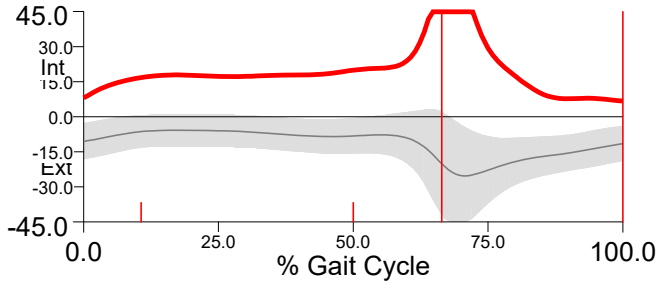
Knee Rotation



Foot Progression



Foot Progression



SHCNC - PreOp

PreOp GPS-MAP Scores

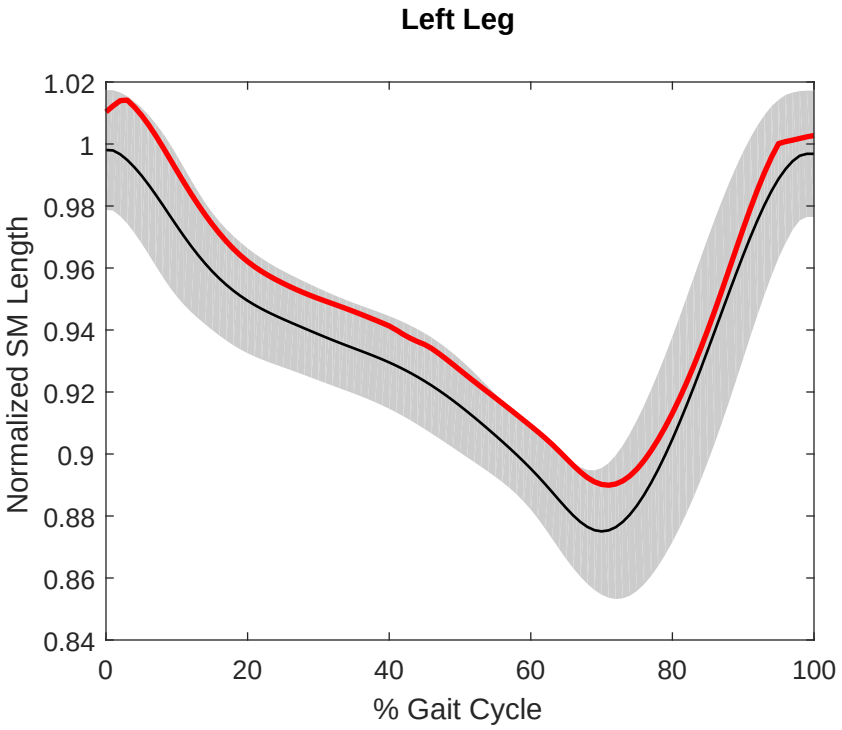
Right

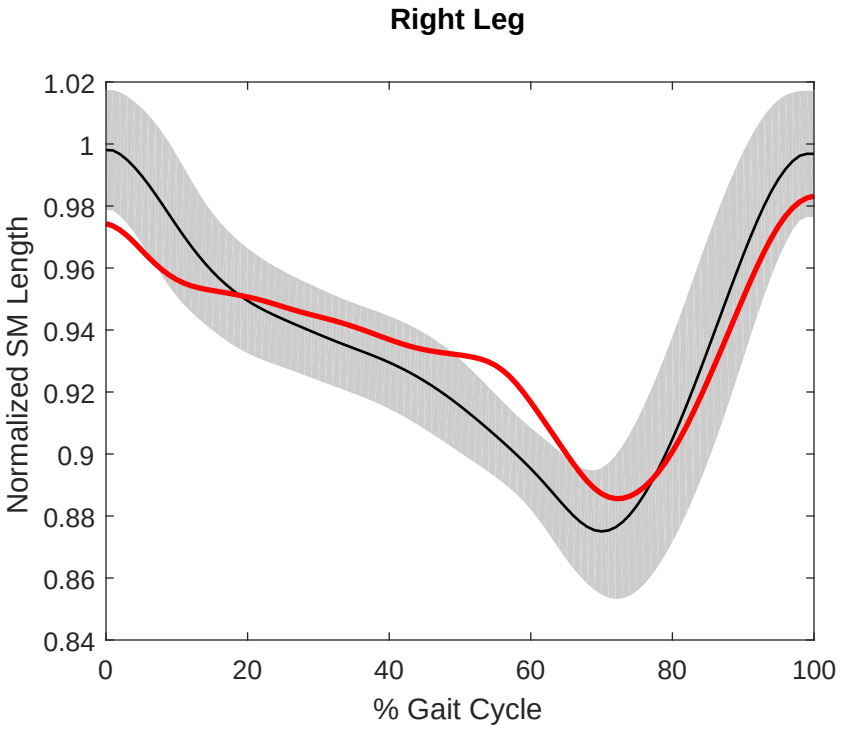
	Bare (deg)	std.dev.
GPS	**17.834**	**1.341**
Pelvic Tilt	5.708	0.587
Pelvic Obliquity	7.622	0.457
Pelvic Rotation	13.294	1.941
Hip Flexion	7.080	1.121
Hip Ab/Adduction	10.263	0.742
Hip Rotation	31.899	1.407
Knee Flexion	7.538	1.582
Ankle Dorsiflexion	24.729	2.072
Foot Progression	27.074	5.374

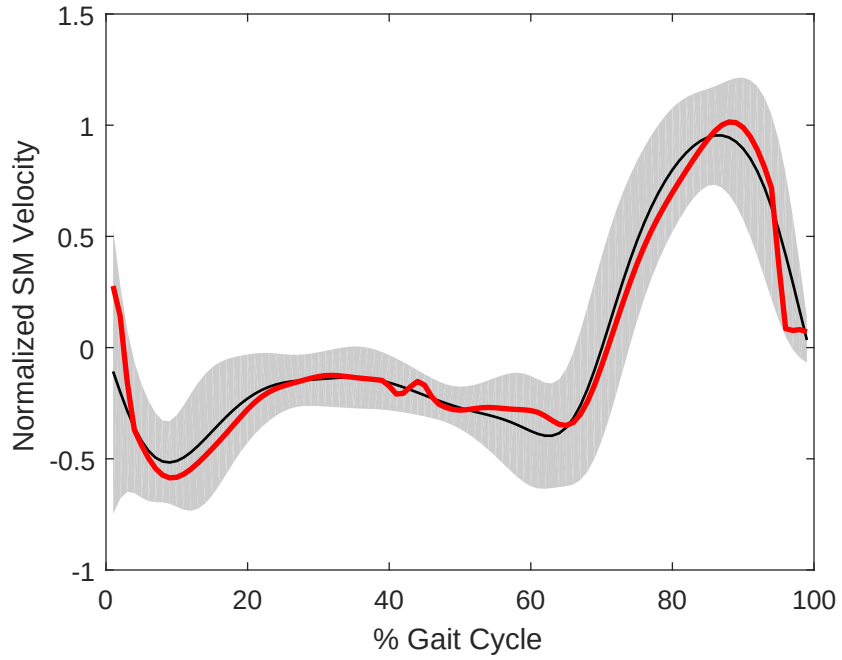
Left

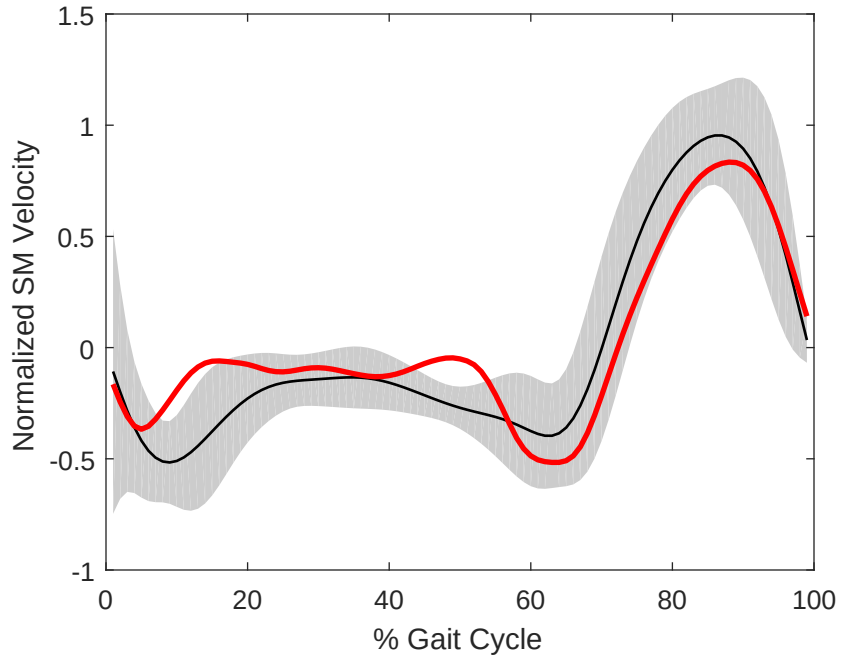
	Bare (deg)	std.dev.
GPS	**8.702**	**0.128**
Pelvic Tilt	5.666	0.434
Pelvic Obliquity	7.655	0.826
Pelvic Rotation	13.146	1.295
Hip Flexion	2.948	0.858
Hip Ab/Adduction	6.958	1.280
Hip Rotation	10.958	1.201
Knee Flexion	8.339	1.884
Ankle Dorsiflexion	5.591	1.345
Foot Progression	11.168	2.961

SHCNC - PreOp


Left Leg
Normalized SM Length
% Gait Cycle
1.02
1
0.98
0.96
0.94
0.92
0.9
0.88
0.86
0.84
0
20
40
60
80
100

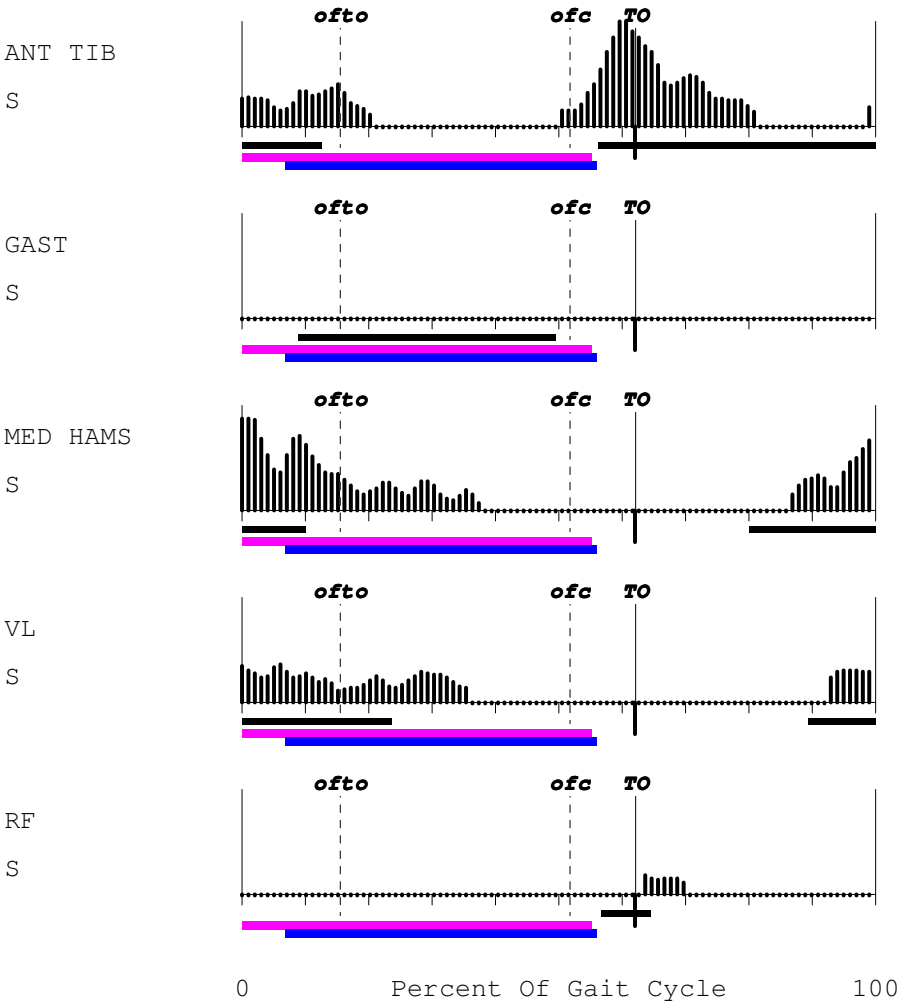

Right Leg
Normalized SM Length
% Gait Cycle
1.02
1
0.98
0.96
0.94
0.92
0.9
0.88
0.86
0.84
0
20
40
60
80
100


Normalized SM Velocity
% Gait Cycle
1.5
1
0.5
0
-0.5
-1
0
20
40
60
80
100


Normalized SM Velocity
% Gait Cycle
1.5
1
0.5
0
-0.5
-1
0
20
40
60
80
100

Shriners Hospitals for Children Northern California Motion Analysis Laboratory

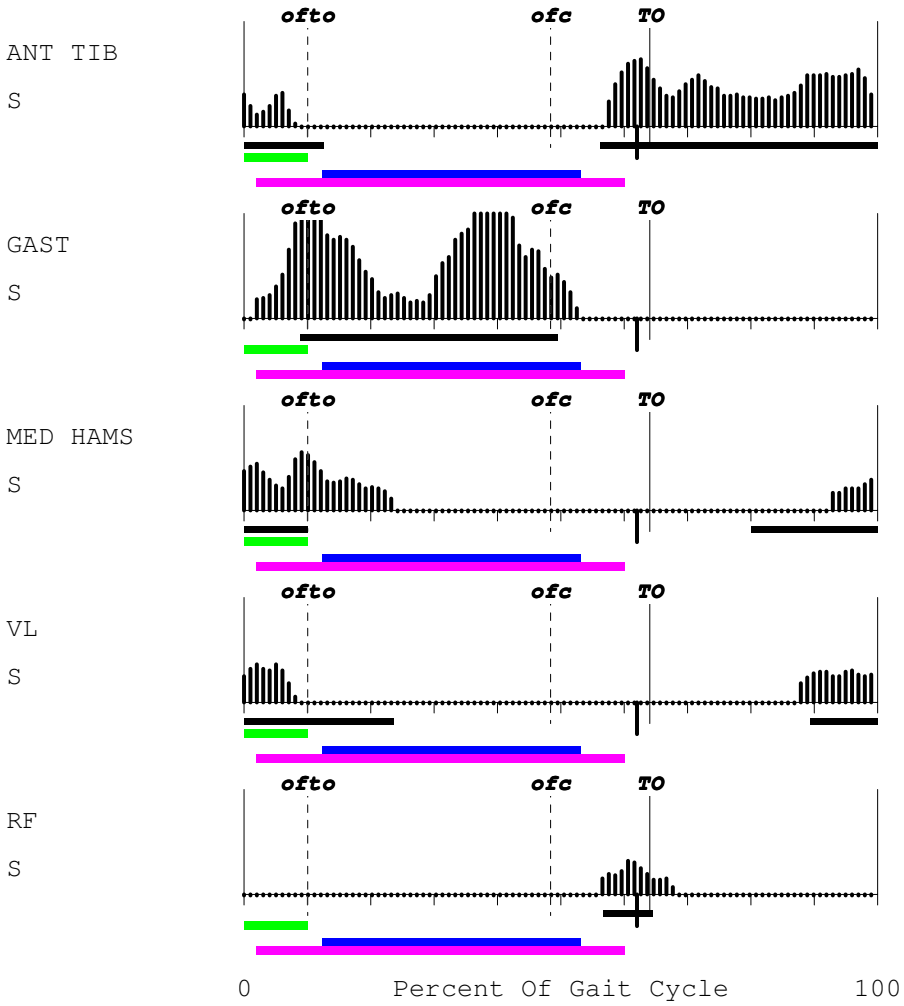
Strides = 5 Side: Right Trial: EMGwalk5 - Walk





Shriners Hospitals for Children Northern California Motion Analysis Laboratory

Strides = 6 Side: Left Trial: EMGwalk5 - Walk

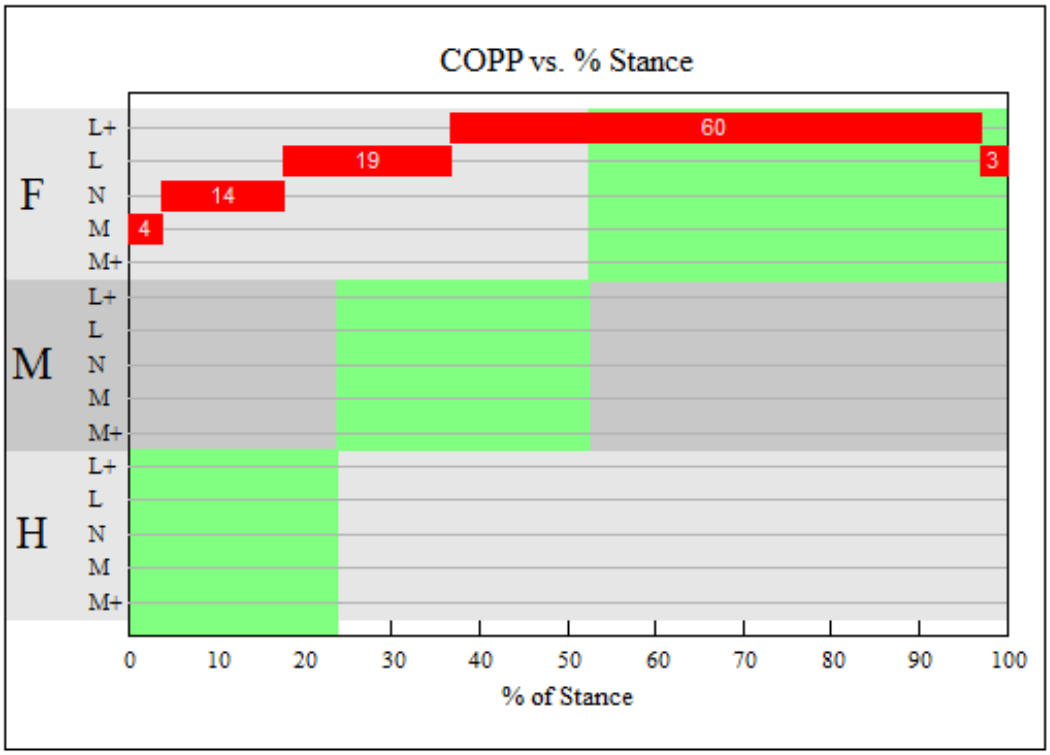


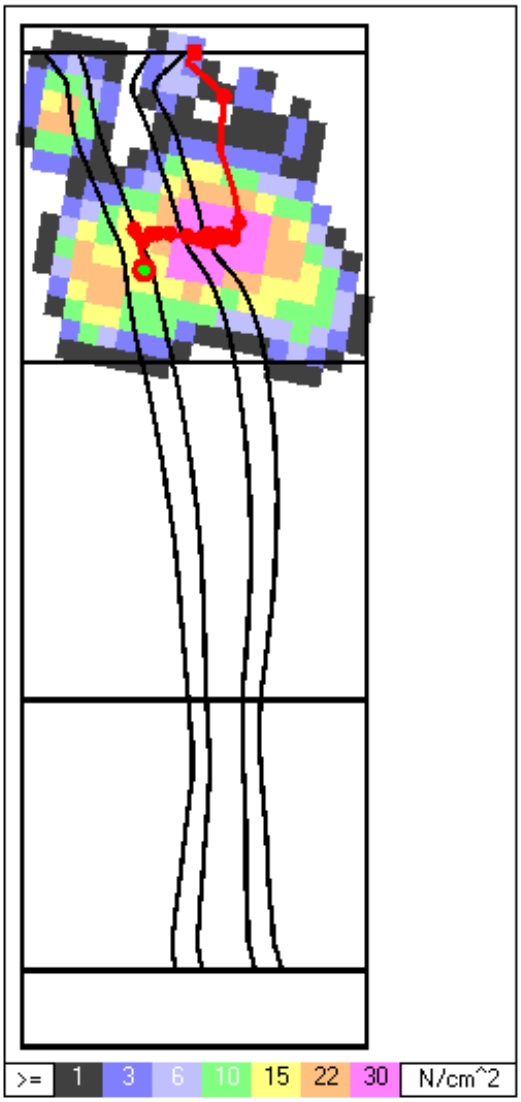


C3D: No C3D file used

C3D Method: ☐ Heel/Toe ☐ Ankle Center/Toe Stance Frame:

RIGHT





	M+	M	N	L	L+	TOTAL
Forefoot	0%	4%	14%	22%	60%	100%
Midfoot	0%	0%	0%	0%	0%	0%
Hindfoot	0%	0%	0%	0%	0%	0%
Total Foot	0%	4%	14%	22%	60%	

Med - Lat

Ant - Post (% of stance)

Initial Sequence: F(100)

Filtered Sequence:

Classification:

Average COPP of Normals
Ant-Post (% of stance)

H(24) - M(29) - F(48)

Foot Progression Angle (deg): 10 Int

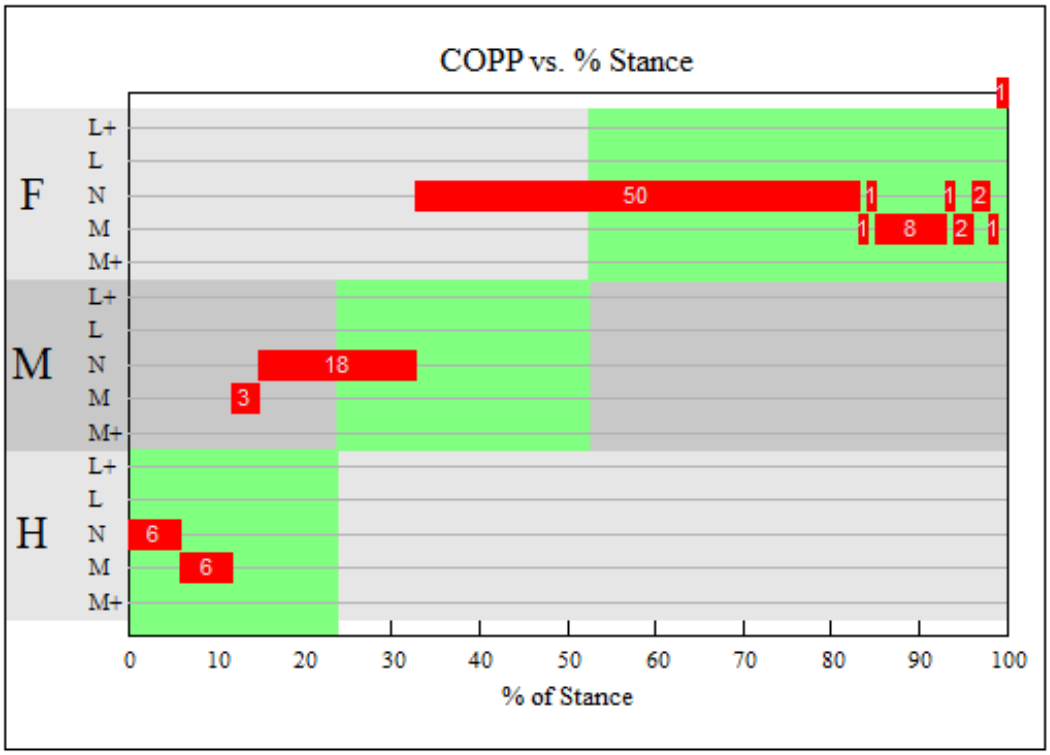
Comments:

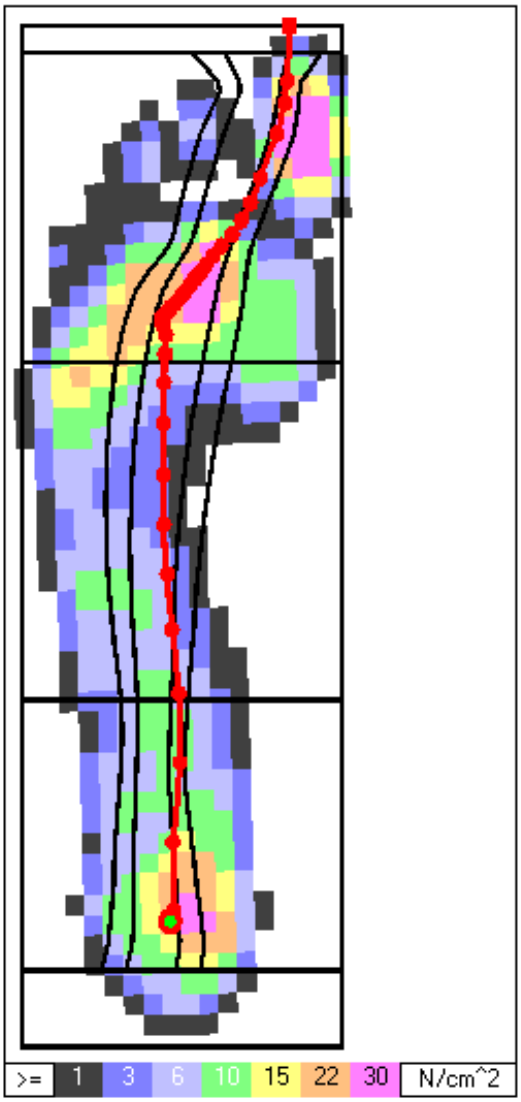
Right Pre Op PEDO

C3D: No C3D file used

C3D Method: ☐ Heel/Toe ☐ Ankle Center/Toe Stance Frame:

LEFT





	M+	M	N	L	L+	TOTAL
Forefoot	0%	12%	54%	0%	0%	66%
Midfoot	0%	3%	18%	0%	0%	21%
Hindfoot	0%	6%	6%	0%	0%	12%
Total Foot	0%	21%	78%	0%	0%	

Med - Lat

Ant - Post (% of stance)

Initial Sequence: H(12)-M(21)-F(67)

Filtered Sequence:

Classification:

Average COPP of Normals
Ant-Post (% of stance)

H(24) - M(29) - F(48)

Foot Progression Angle (deg): 2 Int

Comments:

Left Pre Op PEDO

PODCI Functional Assessment Scores (PODCI AAOS Version 2.0, September 1997)

Parent Responses:

Evaluation Date	Age at Evaluation	Upper Function	Transfers Mobil	Sports Phy Function	Comfort/ Pain	Global Function	Happiness	Relationship
Pre-Op	**10y 1m**	**100**	**97**	**89**	**92**	**94**	**95**	**Father**

Latest Parent Responses Graphed:

